

Gunshot Residue Classification

Accuracy, Interpretability, and Failure Analysis

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DSE 6311—Data Science Capstone

Background

Why GSR Analysis Matters

Problem → misidentifying environmental particles as Gunshot Residue (GSR)

Impact → wrongful convictions & diminished trust in forensic evidence

Our Mission → automated classification tool to:

- Improve consistency
- Enhance forensic reliability in court
- Reduce turnaround time for labs

Stakeholder:

National Institute of Justice (NIJ)

Key Points:

- Traditional SEM/EDS classification relies on expert judgment (slow, expensive, and subjective).
- Environmental confounders like fireworks, brake dust, and industrial materials are high contributors to false positives that may lead to wrongful convictions.

Research Question, Hypothesis & Predictions

Research Question

How accurately and reliably can machine learning models distinguish true gunshot residue particles from chemically similar non-GSR particles, and what are the limits of these models when generalizing to other similar datasets?

Hypotheses

- ML models can effectively classify GSR when multi-element signatures (Pb-Ba-Sb) are present
- Single-element particles will produce substantial classification uncertainty
- Non-linear models (XGBoost, neural networks) will outperform linear models (logistic regression)

Predictions

- Lead, barium, and antimony (and their ratios) will be the most influential features
- Misclassifications will concentrate in barium-heavy non-GSR particles (BaAl, BaCaSi)

Data Sources

Primary: NFI Gunshot Residue Dataset (Netherlands Forensic Institute public GitHub)

- 2,801,667 particles from 210 criminal cases + 63 R&D projects
- SEM/EDS measurements across 90 elemental composition columns
- 4 relational CSV tables; particle data split across 14 files

Validation NIST Gunshot Residue Dataset (National Institute of Standards and Technology)

- Ground-truth particle classifications from known shooter samples
- Used to validate labeling scheme —not merged into training data

Some Caveats:

- NFI data collected over 5 years in the Netherlands may not generalize to US forensic contexts
- Each NIST dataset comes from a single volunteer/range/ammunition type

Data Cleaning & Variable Selection

*Tree model kept all
89 elements

Cleaning

- Concatenated 14 particle CSV files; applied column headers from file 1
- Consolidated 27 original relevance classes → 15 final particle classes per NFI documentation
- Cross-referenced with NIST to assign labels: GSR, Non-GSR, or Ambiguous
- Confirmed zero duplicate observations (stub_id + particle_id)

Variable selection

- 89 elemental features → but only 27 "informative" elements (non-zero in >1% of particles)
- 62 sparse elements dropped* (non-zero in <1% of particles)
- 506,682 ambiguous single-element particles excluded from training

Final dataset

- 2,294,985 particles × 27 elements* + metadata + 1 binary target
- 47% GSR / 53% Non-GSR (no aggressive resampling needed)
- Response variable: Binary GSR label (1 = GSR, 0 = Non-GSR)
 - Chosen because the core forensic question is: "Is this particle gunshot residue or not?"

Modeling Approach

Three models chosen to span the interpretability–performance tradeoff:

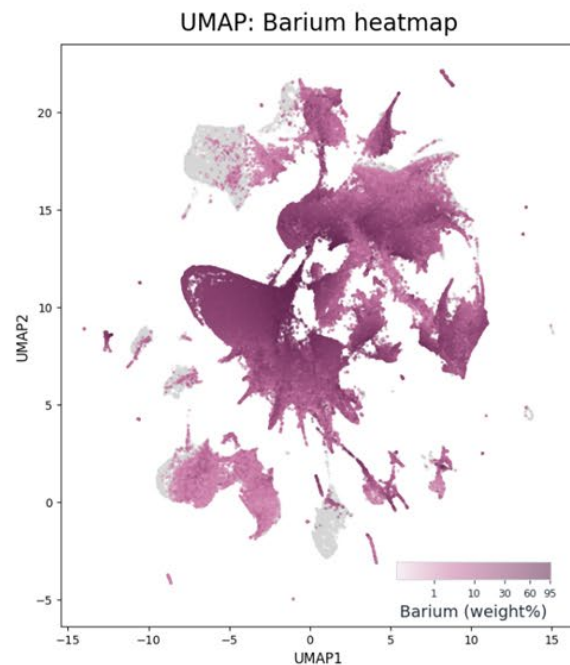
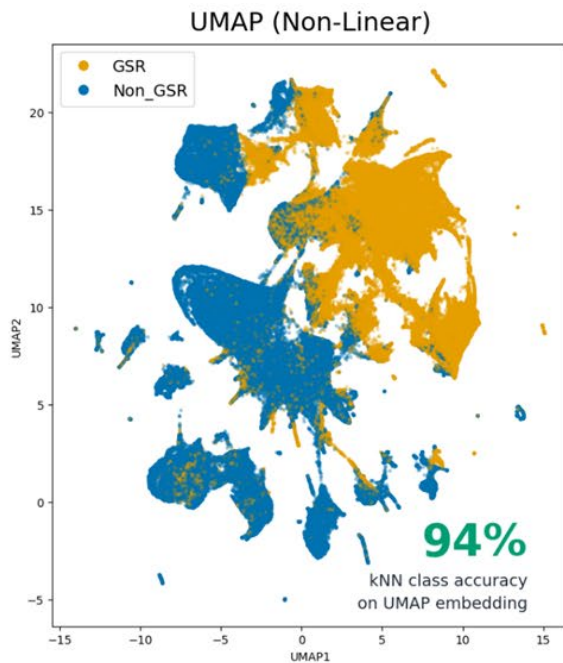
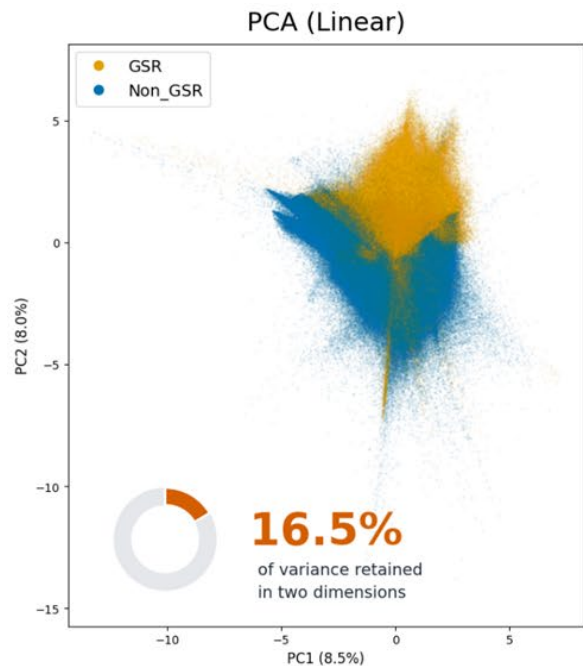
<i>Model</i>	<i>Strengths</i>	<i>Reasoning</i>
Logistic Regression overfitting risk	Interpretable baseline; explainable in court	Transparent coefficients; low
XGBoost importance	Handles sparse, high-dimensional data natively	Non-parametric; built-in feature
Neural Network manual feature engineering not required	Flexible; learns complex interactions	Non-linear;

Tweaks:

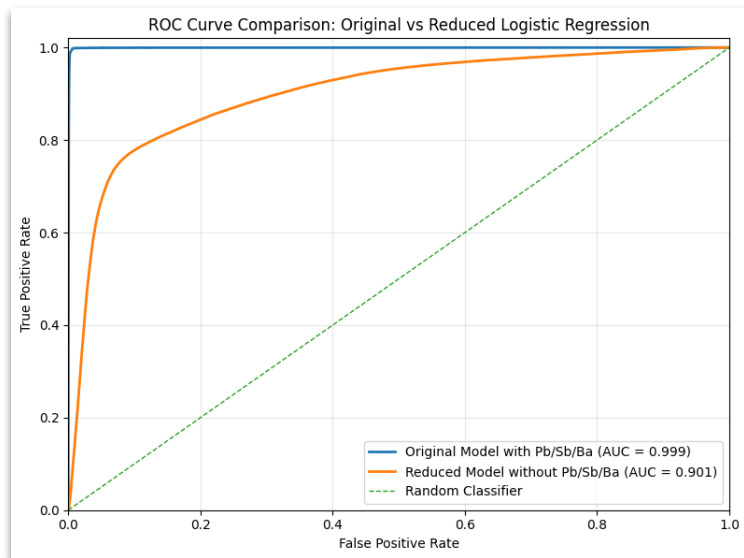
- Group-aware train/test split (GroupShuffleSplit on stub id) to prevent data leakage
- Each model received tailored feature engineering appropriate to its architecture
- Engineered features reviewed for leakage / label proxy (dropped offenders)

Key EDA Insights:

Non-Linear Relationships + Barium Overlap

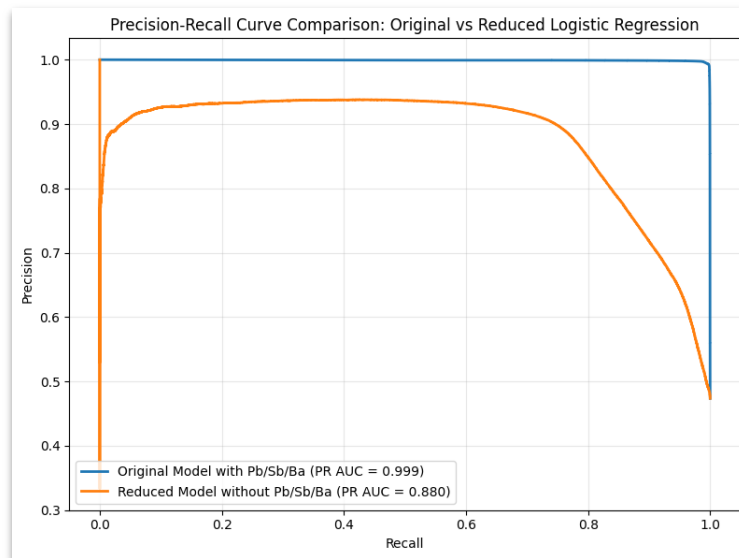


Logistic Regression- Interpretable Baseline Analysis



Approach

- Used proportional elemental features (mass-based %)
- Applied log transformations to reduce skew
- Engineered ratio + interaction features (Pb/BA, Pb x Sb)



Key Results

- Near-perfect ($AUC \approx 1.00$) with full feature set
- Drops to ~ 0.90 when core GSR elements removed

Interpretation

- Relies on core GSR elements (Pb, Sb, Ba)
- Limited ability to capture non-linear relationships

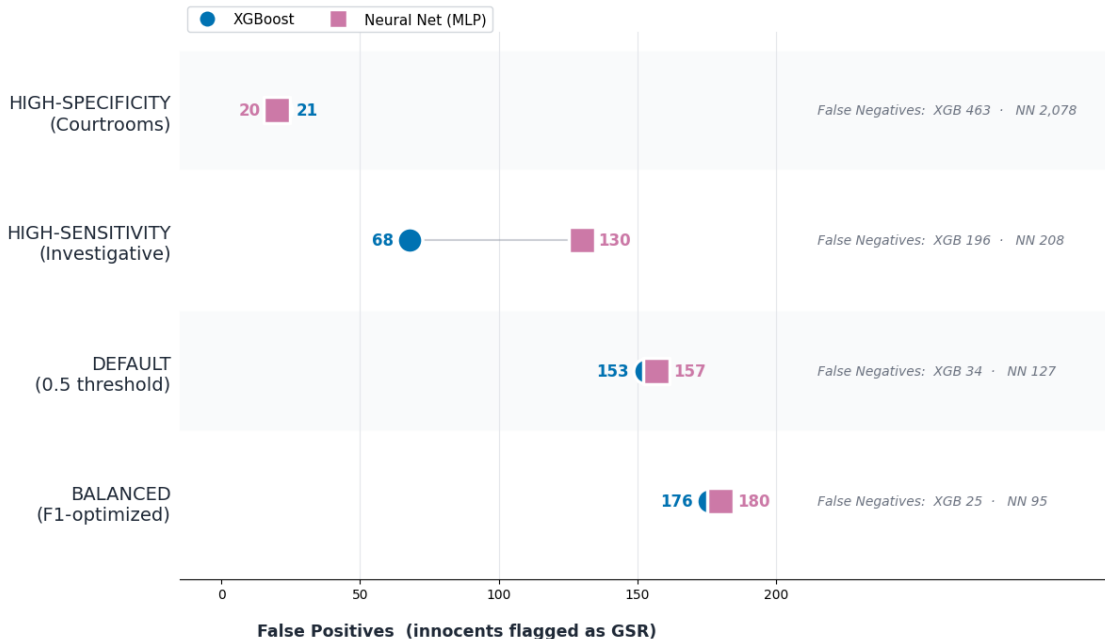
This aligns with our EDA findings –non-linear relationships dominate the data, motivating the use of XGBoost and Neural Networks.

Model Selection: XGBoost or Neural Net?

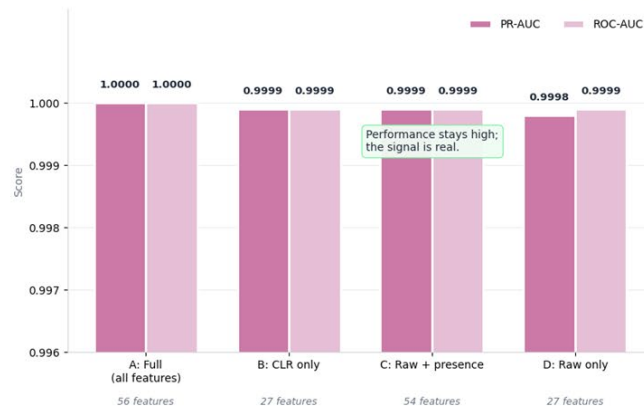
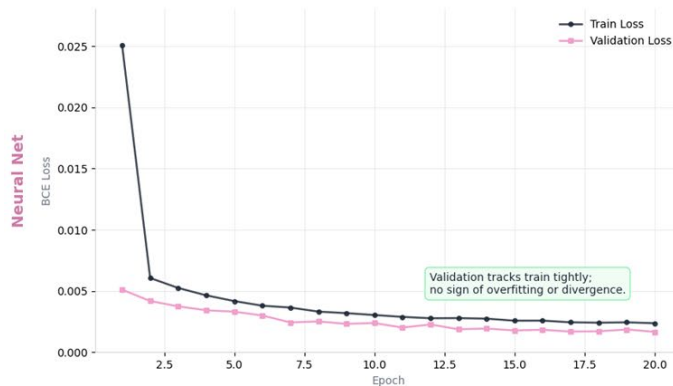
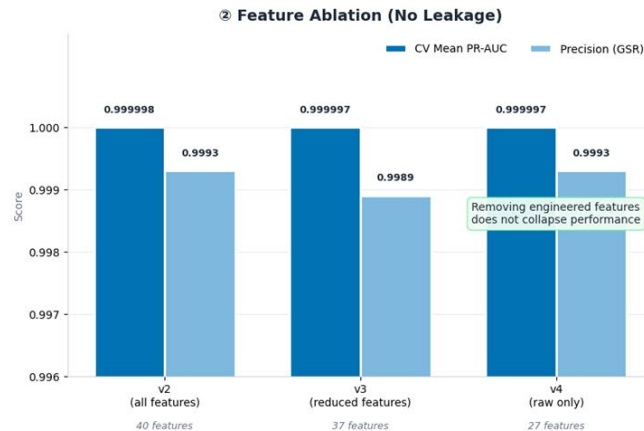
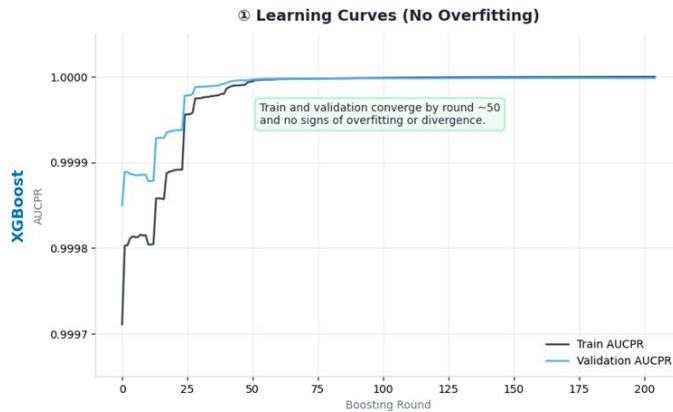
NFI Test Performance

Metric	XGBoost	Neural Net
Features	97	56
Elements	89	27
Accuracy	99.96%	99.93%
ROC-AUC	1.0000	1.0000
PR-AUC	1.0000	1.0000
False Positives	153	157
False Positive Rate	0.07%	0.07%

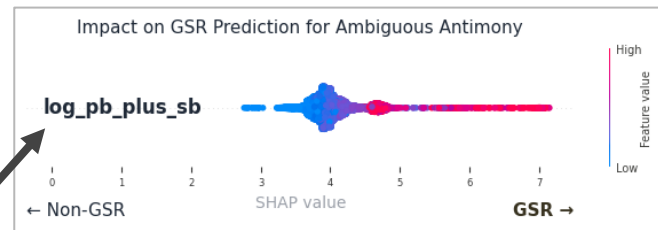
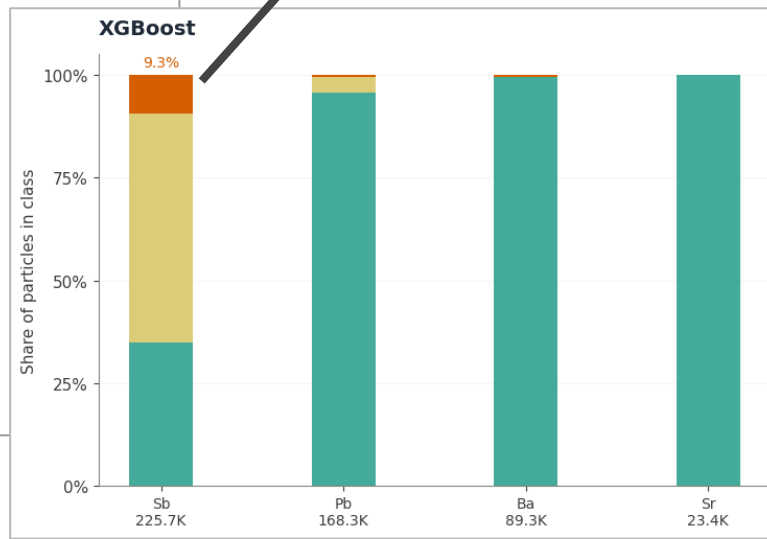
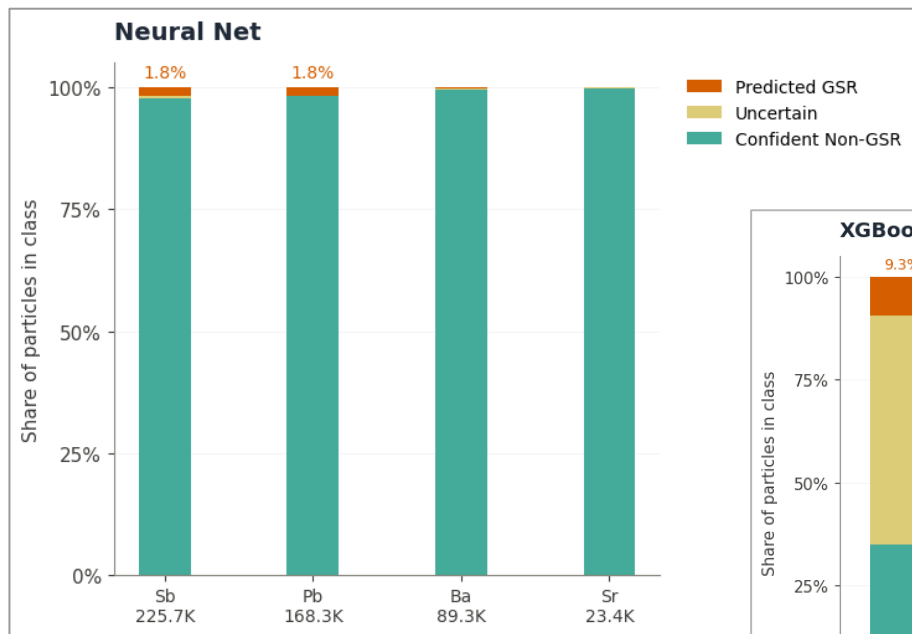
False Positives by Operation Threshold



Failure Mode Checks



Ambiguous Particle Test



Where Errors Happen: False Positive Analysis

- Logistic regression fundamentally cannot learn the non-linear element interactions that define GSR
- XGBoost and the neural network both achieve 99.9%+ accuracy, confirming the task is genuinely separable for non-linear models
- XGBoost has the lowest false positive rate → tree splits are naturally suited to the "if Pb AND Ba AND Sb" logic of GSR classification
- Neural Network is more aggressive at predicting GSR, trading specificity for sensitivity

	Logistic Regression	XGBoost	Neural Network
Total FP	47,618	153	871
FPR	20.4%	0.07%	0.37%
ROC-AUC	0.900	0.9999	1.0000

Where Errors Happen: False Positive Analysis Cont.

- False positives across non-linear models cluster in barium-heavy particle classes (BaAl, BaCaSi)
- These particles likely contain barium from environmental sources (drilling mud, minerals, industrial materials)
- The model sees partial GSR signatures and over-predicts
- This mirrors the exact challenge forensic analysts face: barium is the primary source of false-positive risk in real casework

Class	XGBoost FP	NN FP	Why
BaAl	107	~600+	High barium + trace Pb/Sb contamination
BaCaSi	40	~200+	Barium-heavy environmental mineral
CuZn	3	~50	Brass with trace GSR elements
TiZnGd	3	~20	Rare environmental particle

Testing on Data From a Different Lab: NFI → NIST

- Across datasets (NFI → NIST) XGB and NN catastrophically fail
- The models are not learning generalizable GSR chemistry they're learning NFI-specific measurement patterns
- The real forensic risk: a model deployed in a different lab would produce massive false positive rates
- Key implication: cross-lab validation is essential before any forensic deployment, and the current models are NOT ready for production use outside the NFI context

	XGBoost (NIST)	Neural Network (NIST)
Non-GSR FPR	54.0% (56,881 FP / 105,246)	73.7% (70,328 FP / 95,447)
Shooter GSR rate	not shown separately	63.4%
FP confidence	median 0.9997 (very confident wrong)	median 0.9995 (very confident wrong)
Worst sources	Chevy rear brakes (97%), Ford brakes (80-91%)	Chevy rear brakes (97%), Ford brakes (80-91%)

Recommendations & Next Steps

- Models should be retrained with domain adaptation → the NIST cross-validation proves models learn instrument-specific patterns, not universal GSR chemistry
- Future work should focus on using adversarial domain adaptation to learn instrument-invariant features
- For within-lab deployment (NFI only), the models are production-ready for the binary task, but should include a "defer to expert" flag for single-element ambiguous particles rather than forcing a classification
- The 5 elements missing from NIST (Na, As, Br, W, Hg) and the 14 NIST-only elements not in NFI highlight a standardization gap, a shared minimal element panel across forensic labs would enable better cross-lab model transfer
- MC Dropout or ensemble uncertainty estimation remains valuable not for improving accuracy but for detecting when the model encounters out-of-distribution data (e.g., particles from a different instrument that don't match training patterns)

Citations

Matzen, T., Kukurin, C., van de Wetering, J., Ariëns, S., Bosma, W., Knijnenberg, A., ... & Ypma, R. J. (2022). Objectifying evidence evaluation for gunshot residue comparisons using machine learning on criminal case data. *Forensic science international*, 335, 111293. <https://github.com/NetherlandsForensicInstitute/gunshot-residue>

Ritchie, Nicholas W. M., Renolds, Amy (2021), Automated particle analysis (SEM/EDS) data from samples known to have been exposed to gunshot residue and from samples occasionally mistaken for gunshot residue - like brake dust and fireworks., National Institute of Standards and Technology, <https://doi.org/10.18434/mds2-2476> (Version: 1.0, Accessed 2026-03-21)